

WELL COMPLETION REPORT (SYNTHETIC)

OIL-DGB-013 - Digboi-013

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000145
Field / District	Digboi / Tinsukia
Block	PEL-ASSAM-01
Basin	Assam-Arakan
Formation	Barail (BAR)
Spud / Completion	2024-05-06 / 2024-07-29
Measured / TVD	2722 m / 2694 m
Trajectory	Vertical; inc 1.7 deg; az 51.5 deg
Formation Interval	1702-2401 m
Lithology	Sandstone-shale-coal sequence
Pore Pressure Reference	3257 psi
Fracture Gradient	14.79 ppg
Mud System	PHPA Water-Based Mud
NWIS Well Risk	Moderate (53.9/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	680	Surface
9 5/8 in casing	1578	Intermediate
7 in liner/casing	2395	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2024-06-09 1224 m	Kick	High	Unexpected background gas rise preceded influx. Mitigation: Shut in, verify influx and circulate using approved well-control procedure. Lesson: Treat pit gain plus gas increase as an early influx indicator.
2024-06-15 1416 m	Mud Loss	High	Partial returns observed at shale shaker. Mitigation: Spot graded LCM pill and circulate at reduced rate. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2024-06-18 1502 m	Stuck Pipe	High	Torque and drag increased while ROP declined. Mitigation: Apply approved freeing procedure and avoid excessive overpull. Lesson: Use offset stuck-pipe depth to increase circulation and reaming frequency.
2024-06-19 1537 m	Pressure Spike	High	Circulating pressure climbed without equivalent torque change. Mitigation: Use pressure trend thresholds referenced to depth and flow rate. Lesson: Use depth-matched offset hydraulic baseline for early warning.
2024-06-22 1622 m	Cementing Issue	High	Cement density/returns trend indicated placement uncertainty. Mitigation: Control displacement rate and reconcile pumped/returned volumes. Lesson: Offset loss history should inform cementing hydraulics.
2024-07-16 2328 m	Cementing Issue	High	Displacement pressure became unstable. Mitigation: Verify cement placement using pressure and volume balance. Lesson: Volume imbalance is an important cement-placement warning.
2024-07-25 2591 m	Mud Loss	High	Partial returns observed at shale shaker. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.

Completion Summary

The synthetic well reached 2722 m MD in the Barail interval. The NWIS knowledge record links 13 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.